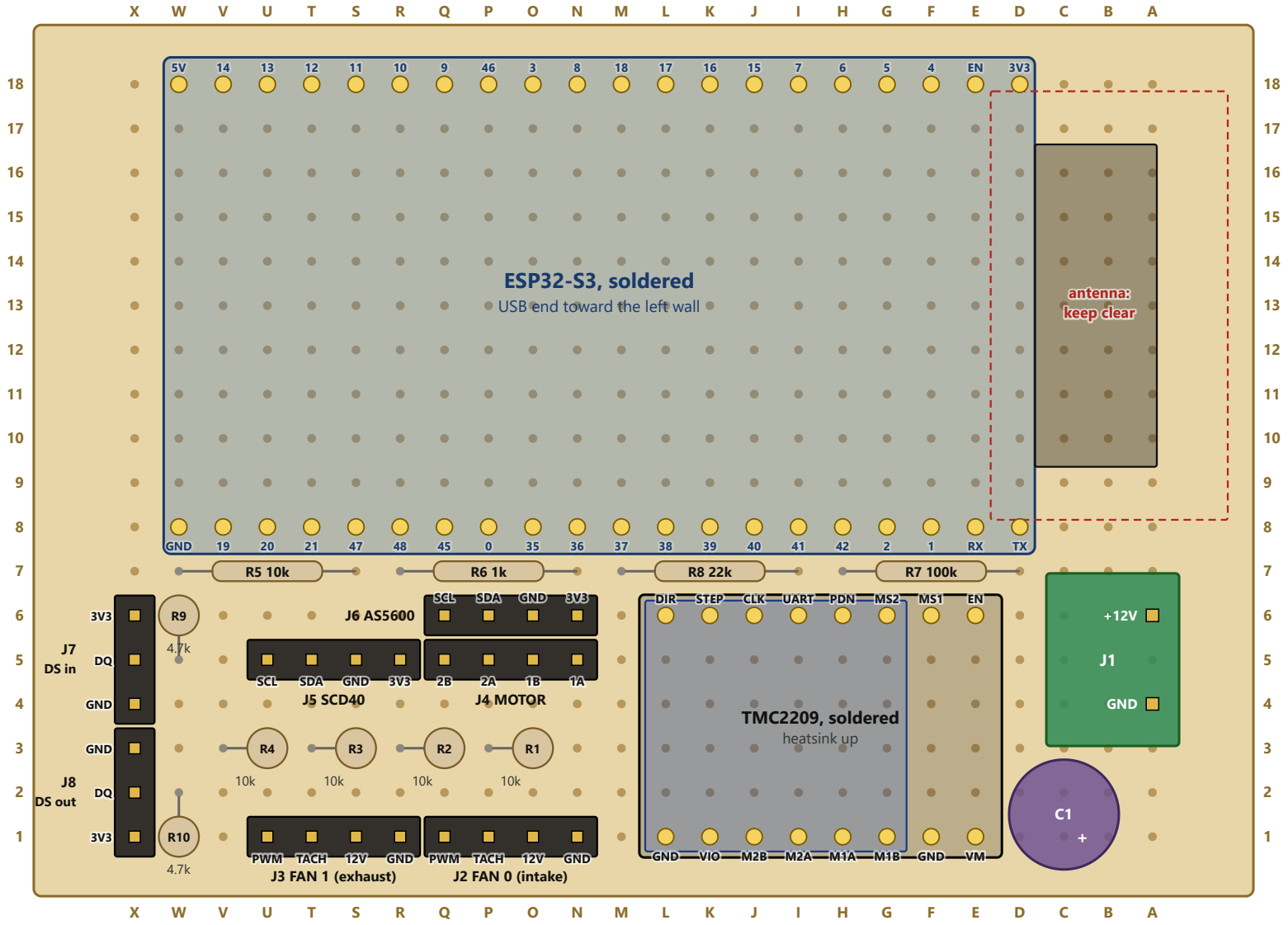
Perfboard build plan (5 x 7 cm protoboard, 24 x 18 holes)

Columns A to X from the right wall, rows 18 (back) to 1 (front), as printed on the board (A18 is its back right corner, seen from above).

Page 1 of 3. Component side, seen from above: window (outdoor) side at the top, room (indoor) side at the bottom



Values: R1-R5 10k, R6 1k, R7 100k, R8 22k, R9 R10 4.7k (1/4 W); C1 100 uF 25 V. R1-R4, R9, R10 stand up; R5-R8 lie flat in row 7, so they pass under the edges of the ESP32 and the TMC2209 (2.6 mm up on their pin spacers).

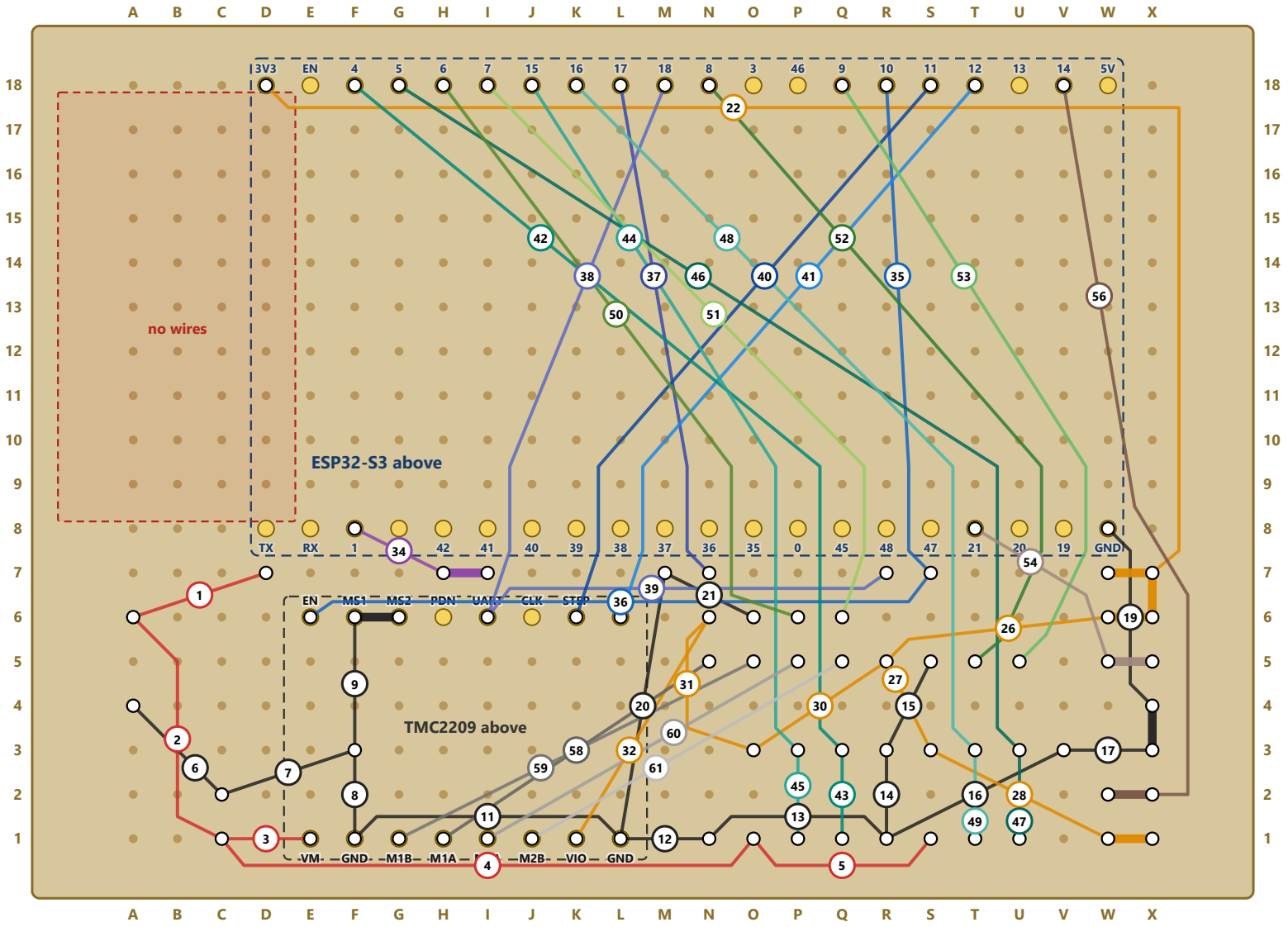
Headers: 1 x 4 and 1 x 3 male, 2.54 mm. The fan plugs (J2, J3) go on with their latch toward row 2; keep row 2 clear there.

AS5600: tie its DIR pin to GND at the module. TMC2209: CLK and PDN stay free (wires 38-39 go to its UART pin; check with a meter that UART is PDN_UART on your module); DIAG (GPIO 13) is not on its pins, so it stays free too.

Solder the parts first, then the ESP32 and the TMC2209: the wires under them must be in place before. Check each net with a meter before power; 12 V on J1 only with the motor plugged in.

Perfboard build plan (5 x 7 cm protoboard, 24 x 18 holes)

Page 2 of 3. Solder side, turned over left to right: A on the left, window side still at the top



Lines are the wires on the underside, coloured by net; the numbers match the list on page 3. Thick short bars are solder bridges between neighbouring holes. Dashed outlines: the ESP32 and the TMC2209, on the other side.

Perfboard build plan (5 x 7 cm protoboard, 24 x 18 holes)

Page 3 of 3. Wires (solder side), with the numbers of page 2

Thin insulated wire (wire-wrap or 26-30 AWG), except 22-24 AWG for the 12 V and GND around the driver (1-14) and the motor (58-61).

1	12V	A6 J1 +12V	to	D7 R7
2	12V	A6 J1 +12V	to	C1 C1 +
3	12V	C1 C1 +	to	E1 TMC VM
4	12V	C1 C1 +	to	O1 J2 12V
5	12V	O1 J2 12V	to	S1 J3 12V
6	GND	A4 J1 GND	to	C2 C1 -
7	GND	C2 C1 -	to	F3 tie point
8	GND	F1 TMC GND	to	F3 tie point
9	GND	F3 tie point	to	F6 TMC MS1
10	GND	F6 TMC MS1	to	G6 TMC MS2 (bridge)
11	GND	F1 TMC GND	to	L1 TMC GND
12	GND	L1 TMC GND	to	N1 J2 GND
13	GND	N1 J2 GND	to	R1 J3 GND
14	GND	R1 J3 GND	to	R3 R2
15	GND	R3 R2	to	S5 J5 GND
16	GND	R1 J3 GND	to	V3 R4
17	GND	V3 R4	to	X3 J8 GND
18	GND	X3 J8 GND	to	X4 J7 GND (bridge)
19	GND	W8 ESP GND	to	X4 J7 GND
20	GND	M7 R8	to	L1 TMC GND
21	GND	O6 J6 GND	to	M7 R8
22	3V3	D18 ESP 3V3	to	X7 tie point
23	3V3	X7 tie point	to	W7 R5 (bridge)
24	3V3	X7 tie point	to	X6 J7 3V3 (bridge)
25	3V3	X6 J7 3V3	to	W6 R9 (bridge)
26	3V3	W6 R9	to	R5 J5 3V3
27	3V3	R5 J5 3V3	to	S3 R3
28	3V3	S3 R3	to	W1 R10
29	3V3	W1 R10	to	X1 J8 3V3 (bridge)
30	3V3	R5 J5 3V3	to	O3 R1
31	3V3	O3 R1	to	N6 J6 3V3
32	3V3	N6 J6 3V3	to	K1 TMC VIO
33	SENSE	H7 R7	to	I7 R8 (bridge)
34	SENSE	H7 R7	to	F8 ESP GPIO 1
35	EN	R18 ESP GPIO 10	to	S7 R5
36	EN	S7 R5	to	E6 TMC EN
37	TX	L18 ESP GPIO 17	to	N7 R6
38	UART	M18 ESP GPIO 18	to	I6 TMC UART
39	UART	R7 R6	to	I6 TMC UART
40	STEP	S18 ESP GPIO 11	to	K6 TMC STEP
41	DIR	T18 ESP GPIO 12	to	L6 TMC DIR
42	PWM0	F18 ESP GPIO 4	to	Q3 R2
43	PWM0	Q3 R2	to	Q1 J2 PWM
44	TACH0	J18 ESP GPIO 15	to	P3 R1
45	TACH0	P3 R1	to	P1 J2 TACH
46	PWM1	G18 ESP GPIO 5	to	U3 R4
47	PWM1	U3 R4	to	U1 J3 PWM
48	TACH1	K18 ESP GPIO 16	to	T3 R3
49	TACH1	T3 R3	to	T1 J3 TACH
50	SDA1	H18 ESP GPIO 6	to	P6 J6 SDA
51	SCL1	I18 ESP GPIO 7	to	Q6 J6 SCL
52	SDA0	N18 ESP GPIO 8	to	T5 J5 SDA
53	SCL0	Q18 ESP GPIO 9	to	U5 J5 SCL
54	DQi	T8 ESP GPIO 21	to	W5 R9
55	DQi	W5 R9	to	X5 J7 DQ (bridge)
56	DQo	V18 ESP GPIO 14	to	X2 J8 DQ
57	DQo	X2 J8 DQ	to	W2 R10 (bridge)
58	M1A	H1 TMC M1A	to	N5 J4 1A
59	M1B	G1 TMC M1B	to	O5 J4 1B
60	M2A	I1 TMC M2A	to	P5 J4 2A
61	M2B	J1 TMC M2B	to	Q5 J4 2B